

Steps	Reason
3. $C \in A$	Given
4. $C \in A \rightarrow C \in B$	Universal instantiation on 1
5. $C \in B \rightarrow C \in C$	Universal instantiation on 2
6. $C \in B$	Modus ponens on 3, 4
7. $C \in A \rightarrow C \in C$	Hypothetical syllogism on 4, 5
8. $\forall x (C \in A \rightarrow x \in C)$	Universal generalization on 7

d) By Defn 3, $A \subseteq C$. $\square$

e) 20) Find two sets A and B such that $A \in B$ and $A \subseteq B$.

f) Solution: $A = \{1\}$. $B = \{1, \{1\}\}$

The subsets of B are: $\emptyset, \{1\}, \{\{1\}\}, \{1, \{1\}\}$.

$\therefore A \in B$ is True and $A \subseteq B$ is True.

g) 21) What is the cardinality of each of these sets?

a) $\{a\}$. a is the only element of the set. $\therefore$ Cardinality is 1

b) $\{\{a\}\}$. $\{a\}$ is the only element of the set. $\therefore$ Cardinality is 1

c) $\{a, \{a\}\}$. The elements of the set are a and $\{a\}$. $\therefore$ Cardinality is 2.

d) $\{a, \{a\}, \{a, \{a\}\}\}$. The elements of the set are $a, \{a\}$ and $\{a, \{a\}\}$. $\therefore$ Cardinality is 3.

e) Find the power set of each of these sets, where a and b are distinct elements.

a) $\{a\}$ $P(\{a\}) = \{\emptyset, \{a\}\}$

b) $\{a, b\}$ $P(\{a, b\}) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

c) $\{\emptyset, \{a\}\}$ $P(\{\emptyset, \{a\}\}) = \{\emptyset, \{\emptyset\}, \{\{a\}\}, \{\emptyset, \{a\}\}\}$

24) Can you conclude that $A = B$ if A and B are two sets with the same power sets?

Solution: Given A and B have the same power sets. By Definition 6 and by Definition 2, the sets of subsets of A and the set of subsets of B contain the same elements. Suppose $A \neq B$. By Defn 2, there exists an element in one set which is not a member of the other set. WLOG, Let $x \in A$ but $x \notin B$.